

Assignment 12: Neural Network and Deep Learning Basics

Image Classification on Fashion-MNIST with a Feedforward Neural Network

1. Introduction

Fashion-MNIST contains 70,000 28x28 grayscale images of clothing across 10 categories (T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot). It is harder than digit-MNIST because several classes share very similar silhouettes (e.g. Shirt vs. T-shirt vs. Coat), making it a good, lightweight testbed for a feedforward neural network and for illustrating where a plain fully-connected model, with no notion of spatial structure, starts to struggle relative to more specialized deep learning architectures.

2. Methodology

Preprocessing: pixel values were rescaled to [0,1]; 6,000 training images were held out for validation; labels were one-hot encoded. Baseline architecture: Flatten ® Dense(128, ReLU) ® Dropout(0.2) ® Dense(10, Softmax), compiled with the Adam optimizer and categorical crossentropy loss, trained for 15 epochs (batch size 64). Experiments: a deeper network (256®128®64 units), a wider network (512 units), the same architecture trained with SGD instead of Adam, and a version with on-the-fly data augmentation (random rotation, translation, zoom, and horizontal flip). Evaluation: accuracy, macro precision/recall/F1, a full classification report, and a confusion matrix were computed on the 10,000-image test set for every configuration.

3. Results

Model	Test Accuracy	Macro F1
Baseline (128 units, Adam)	0.885	0.883
Deeper (256-128-64)	0.875	0.872
Wider (512 units)	0.886	0.884
SGD optimizer (128 units)	0.848	0.844
Augmented baseline	0.800	0.793

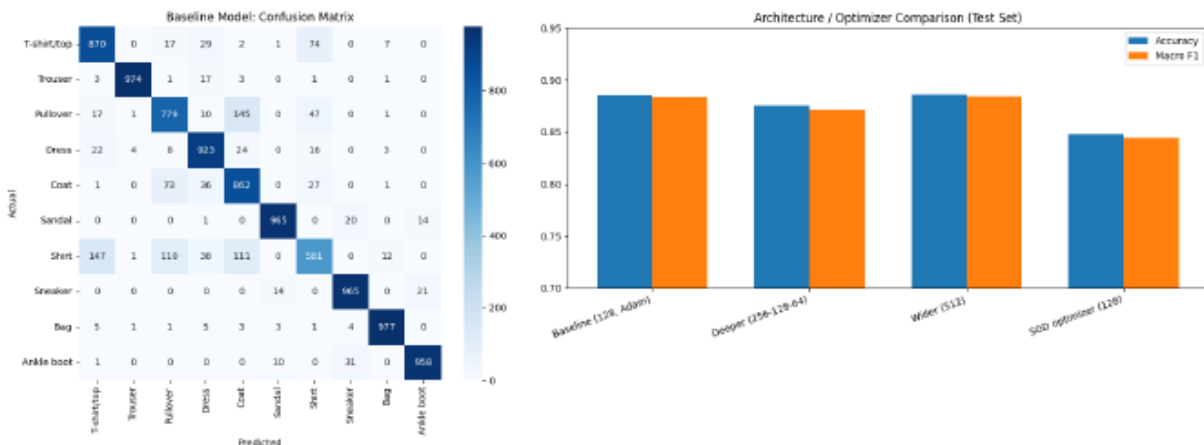


Figure 1 (left): Baseline confusion matrix — confusions concentrate among Shirt / T-shirt / Pullover / Coat. Figure 2 (right): Test accuracy and macro F1 across architectures and optimizers.

The baseline model trained in well under a minute on CPU and reached 88.5% test accuracy. The **wider** network (512 units) matched the baseline as the best result overall; the **deeper** network did not outperform the baseline, since added depth mainly increases optimization difficulty without giving a Dense-only model new spatial features to exploit. Replacing Adam with plain **SGD** clearly underperformed within the same 15-epoch budget. **Data augmentation** (rotation, translation, zoom, flip) reduced accuracy to 80.0%: a fully-connected network has no built-in spatial invariance, so the added input variability mostly acted as noise rather than a useful regularizer — a negative result that would likely reverse with a convolutional architecture and a longer training schedule.

4. Practical Deployment Scenario

Scenario: automated tagging for an online fashion retailer's product catalog, routing new vendor photos into the correct category as they are uploaded. **Scalability:** the trained network is lightweight (a few hundred thousand parameters) and can run high-throughput inference on CPU behind a small container/serverless endpoint. **Real-time needs:** catalog tagging tolerates near-real-time latency (seconds), which fits a simple feedforward model's fast inference well. **Integration:** low-confidence predictions (low max softmax probability) would route to a human reviewer instead of auto-publishing, protecting catalog quality. **Limitation:** real product photos vary far more than Fashion-MNIST (background clutter, color, multiple items per photo), so production use would require a CNN or fine-tuned pretrained vision backbone rather than a plain feedforward network on raw pixels, plus a monitoring/retraining pipeline as new product lines appear.

5. Conclusion

A simple feedforward network reaches ~88% test accuracy on Fashion-MNIST while training in under a minute per configuration on CPU, comfortably above typical raw-pixel classical ML baselines. Widening the network gave a small boost; naive depth increases and unmatched data augmentation did not help a fully-connected architecture. **Future work:** replace the Dense layers with a small CNN (Conv2D + MaxPooling) to better separate visually similar upper-body garments, add learning-rate scheduling and early stopping, retest augmentation with a CNN and a longer schedule, and validate on real (non-benchmark) product photos before considering production deployment.

Link:

<https://colab.research.google.com/drive/17pXNVUChWZtb-x51nTAZ4N71tWXWlewb?usp=sharing>
<https://github.com/Maud20250/Image-Classification-on-Fashion-MNIST-with-a-Feedforward-Neural-Network/upload/main>